

MODULE 3

Hydrogen Energy, Wind Energy, Geothermal Energy

Introduction

Hydrogen is a clean fuel with highest energy content in terms of mass (120.7 MJ/kg). It is possible to use hydrogen directly in IC engines, mix with diesel and CNG and in fuel cells to directly produce electricity. Hydrogen can be used for power generation and as a fuel in automobiles. Hydrogen has been used as a fuel in spacecrafts. When burnt, hydrogen produces water as a by-product. It is, therefore, not only an efficient energy carrier but a clean and environmentally benign fuel as well. Hydrogen can substitute petrol and diesel can and therefore, reduce our dependence on imports.

Hydrogen energy is at present only at the Research, Development and Demonstration (RD&D) stage. The Ministry of New and Renewable Energy is supporting a broad based RD&D projects on different aspects of hydrogen energy technologies including hydrogen production, its storage and utilization for stationary, motive and portable power generation applications using internal combustion engines and fuel cell technologies. The focus of RD&D efforts in this area is directed towards development of new materials, processes, components, sub-systems and systems.

As a result of RD&D efforts made in the area of hydrogen energy, laboratory level prototypes of hydrogen fuelled motorcycles, three wheelers, engine-generator sets, and water/methanol electrolyzers for hydrogen production have been developed. About 15 hydrogen fuelled motorcycles are being demonstrated in the campus of Banaras Hindu University, Varanasi. Hydrogen catalytic combustion cookers have been developed. Facilities for dispensing hydrogen blended compressed natural gas fuel have been set up in Faridabad and Delhi.

The Ministry of Science and Technology, CSIR Laboratories, Ministry of Petroleum and Natural Gas, Defence Research & Development Organizations, Indian Space Research Organization, Oil & Gas companies, Department of Atomic Energy and private sector automobile companies are also involved in the research, development and demonstration programme related to hydrogen.

Hydrogen is available in abundance in water, biomass and hydrocarbons. However, it is not found as an independent gas in the atmosphere.

At present, hydrogen is being commercially produced in the country in the fertilizer, petroleum refining and chemical industries. In addition, hydrogen is also produced as a by-product in chlor-alkali industries. Some limited hydrogen is also produced through electrolysis for commercial use. The estimated hydrogen production and consumption in the country during 2007-08 as per a study undertaken by the University of Petroleum and Energy Studies (UPES), Dehradun are as given below:

Sector	Estimated Production during 2007-08 (Million Tonnes/Year)	Utilization during 2007-08 (Million Tonnes / Year)
Fertilizer Industry	1.99	1.99 (Captive Use)
Petroleum Refineries	1.69	1.462 (Captive use)
Chlor Alkali Industry	0.073	0.064
Total	3.753	3.516

Hydrogen Production Technologies

Energy storage using hydrogen-electrolysis systems increasing the share of wind in UK electricity generation is likely to introduce a problem of power generation intermittency. Electricity cannot be stored on a large scale but it has been proposed that the excess could be stored by producing hydrogen through electrolysis. 12 The electrolyzers would be operated intermittently, which could cause efficiency reductions and introduce heat management and safety issues (Sherif et al. 2005). If the electrolyser temperature is too low, due to a low production rate, then the efficiency of the electrolyser will reduce. A low production rate could also allow hydrogen and oxygen to permeate through the electrolyte and come into contact, reducing the efficiency and possibly causing a fire. Alkaline electrolyzers are particularly sensitive to these problems but PEM electrolyzers are also affected to a lesser extent. Hydrogen would be produced predominantly using off-peak electricity at times of low demand. While this would decrease the running costs, it would increase the capital investment costs of the electrolyzers because production would be substantially below capacity. Any analysis of intermittent production would have to carefully consider the electrolyser utilization factor. 3.4.2 Thermochemical cracking of water It was mentioned above that SOE electrolyzers operating at higher

temperatures achieve greater hydrogen production efficiencies. Hundreds of other high-temperature chemical cycles have been identified for producing hydrogen (Rosen 2010). Studies have primarily concentrated on two technologies to produce the energy for these thermochemical cycles. Firstly, high-temperature nuclear reactors could supply both electricity and high-temperature waste heat to an adjacent production plant (Utgikar and Thiesen 2006; Lubis et al. 2010). Secondly, concentrated solar power (CSP) plants could produce both electricity and the required temperatures in a central tower for hydrogen production (Felder and Meier 2008; Coelho et al. 2010). The USA has funded substantial research into fourth-generation high-temperature helium reactors for hydrogen production under the DOE Nuclear Hydrogen Initiative program (Bartels et al. 2010) but no plants have been demonstrated on an industrial scale. The high uncertainty in the capital costs of such a system are reflected by the wide range of estimates in Figure 2. Cost evaluation tools are now being developed (e.g. Khamis and Malshe 2010) so more authoritative estimates might become available in future. The different technology energy efficiencies (Figure 3) reflect the different efficiency measurements that are used in the literature. The higher estimate only considers the hydrogen production plant (Steward et al. 2008) while the lower estimate represents the entire nuclear reactor and hydrogen production system (Khamis and Malshe 2010). For the hydrogen production process alone, the energy efficiency should exceed 90% in the future. The costs of CSP are even more speculative. Most authoritative studies are relatively old (e.g. Glatzmaier et al. 1998) and both the costs and efficiencies are uncertain at the moment due to the early stage of development of the technologies (Steinfeld 2005), so no data is presented here. Both costs and efficiencies would be strongly influenced by the location of the CSP plant due to the variations in solar radiation across the planet.

Advantages and Disadvantages of Hydrogen Energy

Advantages of Hydrogen Energy

1. It's a renewable energy source and bountiful in supply

Hydrogen is a rich source of energy for many reasons; the main being that it's bountiful in supply. While it may take a lot of resources to harness it, no other energy source is infinite as hydrogen. That, essentially, means there is no possibility of it running out like other sources of energy.

2. It practically a clean energy source

When hydrogen is burnt to produce fuel, the by products are totally safe, which means, they have no known side effects. Aeronautical companies actually use hydrogen as a source of drinking water. After hydrogen is utilized, it is normally converted to drinking water for astronauts on ship or space stations.

3. Hydrogen energy is non-toxic

This means that it does not cause any harm or destruction to human health. This aspect makes it preferred compared to other sources of fuel like nuclear energy, natural gas, which are extremely hazardous or daunting to harness safely. It also allows hydrogen to be used in places where other forms of fuel may not be allowed.

4. It's far more efficient than other sources of energy

Hydrogen is solidly efficient energy type since it has the ability to convey a lot of energy for every pound of fuel. This categorically means that an automobile that utilizes hydrogen energy will travel more miles than one with an equal amount of gasoline.

5. Used for powering space ships

Hydrogen energy's efficiency and power makes it an ideal fuel source for spaceships. Its power is so high that it's able to quickly rocket spaceships to exploration missions. It's also the safest form of energy to perform such an energy-intensive task. Hydrogen energy is in fact 3 times more potent than gasoline and other fossil-based sources of fuel. This ideally means that you need less hydrogen to complete an enormous task.

It also offers motive power for airplanes, boats, cars, and both portable and stationary fuel cell applications. The downside to using hydrogen in cars is that it's practically difficult to store in cryogenic or high-pressure tanks.

Disadvantages of Hydrogen Energy

While hydrogen energy has a lot of admirable benefits, it's not really the outright preferable, clean and cheap energy source for most governments and companies. In gaseous state, it's quite volatile. While its volatility gives it an edge over energy sources in terms of accomplishing numerous tasks, it equally renders it risky to use and work around. Some of the disadvantages of hydrogen energy include:

1. Hydrogen energy is expensive

Electrolysis and steam reforming, the two main processes of hydrogen extraction are extremely expensive. This is the real reason it's not heavily used across the world. Today, hydrogen energy is chiefly used to power most hybrid vehicles. A lot of research and innovation is required to discover cheap and sustainable ways to harness this form of energy. Until then, hydrogen energy would remain exclusively for the rich.

2. Storage complications

One of hydrogen properties is that it has a lower density. In fact, it is a lot less dense than gasoline. This means that it has to be compressed to liquid state and stored the same way at lower temperatures to guarantee its effectiveness and efficiency as an energy source. This reason also explains why hydrogen must at all times be stored and transported under high pressure, which is why transportation and common use is far from feasible.

3. It's not the safest source of energy

The power of hydrogen should not be underestimated at all. Although gasoline is a little more dangerous than hydrogen, hydrogen is hugely flammable and frequently makes headlines for its potential dangers. Compared to gas, hydrogen lacks smell, which makes any leak detection almost impossible. To detect leaks, one must install sensors.

4. Tricky to move around

It's a daunting task to transport hydrogen brilliantly due to its lightness. Oil can be transported safely because it's mostly pushed through pipes. Coal can conveniently be transported in dump trucks. Hydrogen also presents challenges when considering moving it in large quantities, which is why it's mostly only transported in small batches.

5. Hydrogen energy cannot sustain the population

Despite the fact that hydrogen is bountiful in supply, the cost of harnessing it limits extensive utilization. As you realize, it's quite challenging to disrupt the status quo. Energy from fossil fuels still rules the world. There is also no framework put in place to ensure cheap and sustainable hydrogen energy for the normal car owner in the future. Even if hydrogen were to become cheap right now, it would take years to become the most used source of energy since vehicles themselves

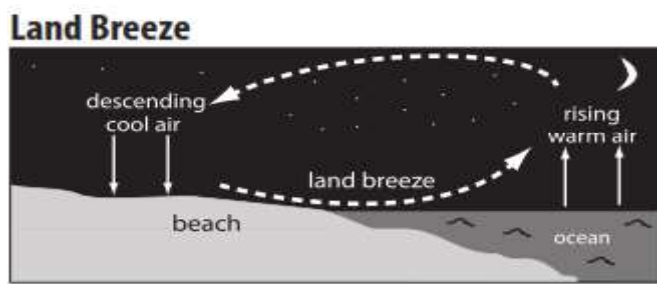
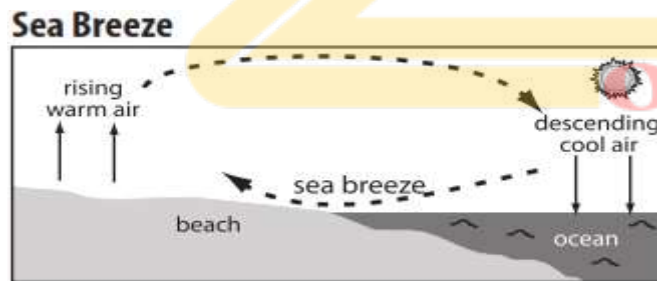
and service stations would need to be customized to conform to hydrogen requirements. This would require massive capital outlay.

WIND ENERGY

Introduction:

Winds are essentially created by the solar heating of the atmosphere. Several attempts have been made since 1940 to use wind to generate electric energy and development is still going on. However, techno-economic feasibility has yet to be satisfactorily established. Wind as a power source is attractive because it is plentiful, inexhaustible and non-polluting. Further, it does not impose extra heat burden on the environment. Unfortunately, it is non-steady and undependable. Control equipment has been devised to start the wind power plant whenever the wind speed reaches 30km/h. Methods have also been found to generate constant frequency power with varying wind speeds and consequently varying speeds of wind mill propellers. Wind power may prove practical for small power needs in isolated sites. But for maximum flexibility, it should be used in conjunction with other methods of power generation to ensure continuity. For a rotor of 17m diameter and a velocity of 48 km/h the theoretical power is 265kW and the practical would be roughly half of this value.

There are some distinctive energy end-use features of wind power systems: Most wind power sites are in remote rural, island or marine areas. Rural grid systems are likely to be 'weak' in these areas, since they carry relatively low voltage supplies (e.g. 33kV).



Wind energy:

Can be economically used for the generation of electrical energy. Winds are caused due to Heating and cooling of the main atmosphere which generates convection currents. The rotation of the earth with respect to atmosphere, and its motion around the sun. The potential of wind energy is abundant. 1.6×10^7 MW. (Same order of present energy consumption) Wind mill is drives generator to produce electricity.

Water pumping for irrigation and drinking water Required Wind speed range is 8 to 36Km per hour In India, coastal areas of Saurashtra, western Rajasthan and some parts of central India.

Wind Direction:

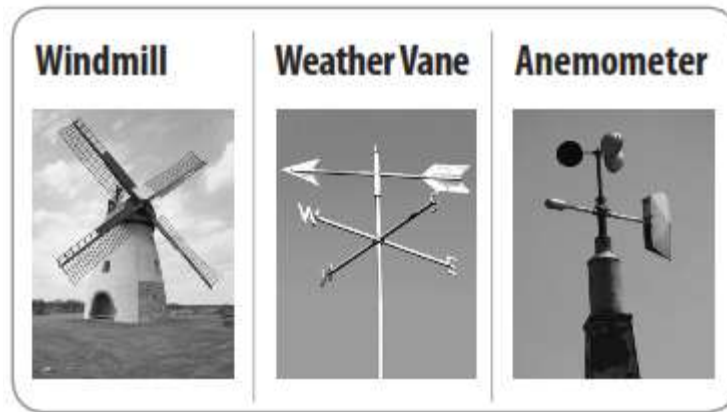
A weather vane, or wind vane, is used to show the direction of the wind. A wind vane points toward the source of the wind. Wind direction is reported as the direction from which the wind blows, not the direction toward which the wind moves. A north wind blows from the north toward the south.

Wind Speed:

It is important in many cases to know how fast the wind is blowing. Wind speed can be measured using a wind gauge or anemometer. One type of anemometer is a device with three arms that spin on top of a shaft. Each arm has a cup on its end. The cups catch the wind and spin the shaft. The harder the wind blows, the faster the shaft spins. A device inside counts the number of rotations per minute and converts that figure into miles per hour. A display on the anemometer shows the speed of the wind.

History of Wind Machines:

Since ancient times, people have harnessed the wind's energy. Over 5,000 years ago, the ancient Egyptians used the wind to sail ships on the Nile River. Later, people built windmills to grind wheat and other grains. The early windmills looked like paddle wheels. Centuries later, the people in Holland improved the windmill. They gave it propeller type blades, still made with sails. Holland is famous for its windmills. In this country, the colonists used windmills to grind wheat and corn, to pump water, and to cut wood at sawmills. Today, people occasionally use windmills to grind grain and pump water, but they also use modern wind turbines to make electricity



Date	Typical Capacity	Typical Blade Length	Typical Technology
Mid 1990s	400-500 kW	15-25 m	Fixed rotational speed and fixed blade pitch angle
2000	1000 kW	25-35 m	Dual rotational speed and fixed blade pitch angle
Today	2000-3000 kW	35-45 m	Variable rotational speed and variable blade pitch angle
Within 5 years	3000-7000 kW	45-60 m	

Today's Wind Turbines:

Like old-fashioned windmills, today's wind turbines use blades to capture the wind's kinetic energy. Wind turbines work because they slow down the speed of the wind. When the wind blows, it pushes against the blades of the wind turbine, making them spin. They power a generator to produce electricity. Most wind turbines have the same basic parts: blades, shafts, gears, a generator, and a cable. (Some turbines do not have gearboxes.)

These parts work together to convert the wind's energy into electricity.

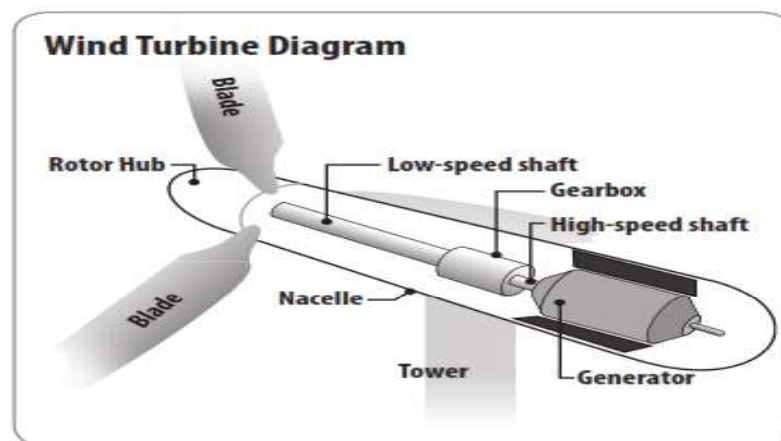
1. The wind blows and pushes against the blades on top of the tower, making them spin.
2. The turbine blades are connected to a low-speed drive shaft. When the blades spin, the shaft turns. The shaft is connected to a gearbox. The gears in the gearbox increase the speed of the spinning motion on a high-speed drive shaft.
3. The high-speed drive shaft is connected to a generator. As the shaft turns inside the generator, it produces electricity.
4. The electricity is sent through a cable down the turbine tower to a transmission line.

The amount of electricity that a turbine produces depends on its size and the speed of the wind. Wind turbines come in many different sizes. A small turbine may power one home. Large wind turbines can produce enough electricity to power up to 1,000 homes. Large turbines are sometimes grouped together to provide power to the electricity grid. The grid is the network of power lines connected together across the entire country

Wind Power Plants:

Wind power plants, or wind farms, are clusters of wind turbines used to produce electricity. A wind farm usually has dozens of wind turbines scattered over a large area. Choosing the location of a wind farm is known as siting a wind farm. The wind speed and direction must be studied to determine where to put the turbines. As a rule, wind speed increases with height, as well as over open areas with no windbreaks.

Turbines are usually built in rows facing into the prevailing wind. Placing turbines too far apart wastes space. If turbines are too close together, they block each other's wind. The site must have strong, steady winds. Scientists measure the winds in an area for several years before choosing a site. The best sites for wind farms are on hilltops, on the open plains, through mountain passes, and near the coasts of oceans or large lakes. The wind blows stronger and steadier over water than over land. There are no obstacles on the water to block the wind. There is a lot of wind energy available offshore. Offshore wind farms are built in the shallow waters off the coast of major lakes and oceans. Offshore turbines produce more electricity than turbines on land, but they cost more to build and operate. The first offshore wind farm in the United States, off the coast of Massachusetts, was approved in April 2011. Construction is expected to begin in 2013.



Wind Production

Every year, wind produces only a small amount of the electricity this country uses, but the amount is growing every year. One reason wind farms don't produce more electricity is that they can only run when the wind is blowing at certain speeds. On Midwestern wind farms, the wind is optimum for producing electricity between 65 and 90 percent of the time.



Environmental Impacts

In some areas, people worry about the birds and bats that may be injured by wind turbines. Some people believe wind turbines produce a lot of sound, and some think turbines affect their view of the landscape. On the other hand, wind is a clean, renewable energy source that produces no air pollution. And wind is free to use. Wind power is not the perfect answer to our electricity needs, but it is a valuable part of the solution.

Power in a wind stream

A wind stream has total power given by $P_t = \dot{m} \cdot (K.E.w) = 0.5 \dot{m} \cdot V_i^2$

Where, $\dot{m}$ = mass flow rate of air, kg/s

V_i = incoming wind velocity, m/s

Air mass flow rate is given by

$$\dot{m} = \rho A V_i$$

Where, ρ = Density of incoming wind, $\text{kg/m}^3 = 1.226 \text{ kg/m}^3$ at 1 atm, 15°C

A = Cross-sectional area of wind stream, m^2

Substituting the above and accounting for the constants, we arrive at the following:

$$P_w = 0.5 \rho R^3 V_w^3 C_P(\lambda, \beta)$$

Where,

P_w = extracted power from the wind,

ρ = air density, (approximately 1.2 kg/m^3 at 20°C at sea level)

R = blade radius (in m), (it varies between 40-60 m)

V_w = wind velocity (m/s) (velocity can be controlled between 3 to 30 m/s)

C_p = the power coefficient which is a function of both tip speed ratio (λ), and blade pitch angle, (β) (deg.)

Power coefficient (C_p) is defined as the ratio of the output power produced to the power available in the wind.

Betz Limit

Betz limit is the theoretical limit assigned to efficiency of a wind turbine. It states that no turbine can convert more than 59.3 % of wind kinetic energy into shaft mechanical energy. Thus the value of C is limited to Betz limit. For a well designed turbine the efficiency lies in the range of 35-45 %.

Wind Turbines

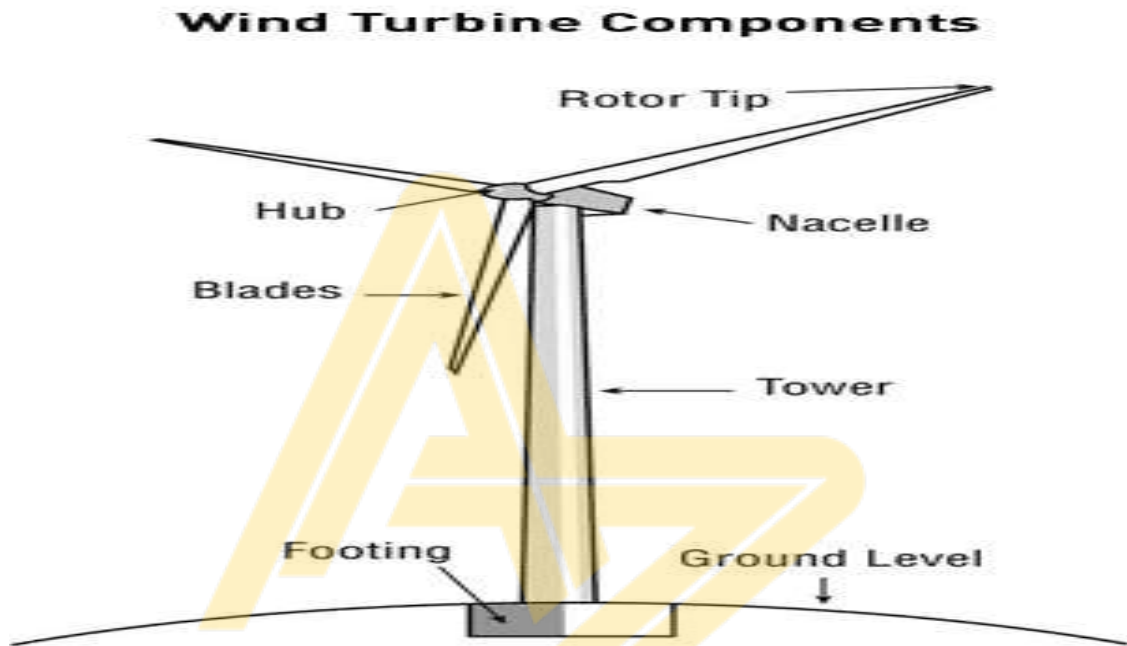
A wind turbine is a rotating machine which converts the wind kinetic energy into mechanical energy. If the mechanical energy is then converted to electricity, the machine is called a wind generator, wind turbine. Wind turbines can be separated into two types based by the axis in which the turbine rotates as Horizontal Axis Wind Turbines and Vertical Axis Wind Turbines. The former are more commonly used due to several inherent advantages, the latter being used in small scale.



Wind Turbine Generator units

Turbine subsystems include:

- Rotors which convert wind energy into mechanical energy of the shaft ;
- Nacelle (enclosure) which contains all the conversion equipment, generator ,gear shaftetc.
- Tower, to increase the height of the turbine systems so that higher wind speeds arecaptured.
- Control equipment, Cables and other Civil works.



Horizontal Axis Wind Turbines (HAWTs)

Horizontal-axis wind turbines (HAWT) get their name from the fact that their axis of rotation is horizontal. They have the main rotor shaft and electrical generator at the top of a tower, and are pointed into the wind. The variability of wind distribution and speed brings up the requirement of a gear system connected to the rotor and the generator. The gear system enables a constant speed of rotation to the generator thus enabling constant frequency generation. Turbine blades are made stiff in order to prevent the blades from being pushed into the tower by high winds. Downwind machines have also been built, as they no longer require a yaw mechanism to keep them facing the wind, and also because in high winds the blades can turn out of the wind thereby increasing drag and coming to a stop. Most of the HAWTs' are upwind as downwind systems cause regular turbulence which may lead to fatigue.

HAWT advantages

- Variable blade pitch, which gives the turbine blades the optimum angle of attack. Changing the angle of attack provides greater control over power generated and enables maximum efficiency.
- As wind energy increases with height, the tall tower in the HAWT gives access to higher wind speed. In some cases increase of even 10m height leads to increase in wind speed by 20 %
- In HAWTs' the blades move horizontally that is perpendicular to the wind and hence have minimum drag and they receive power throughout the rotation.

HAWT disadvantages

- Due to inherent large structures, construction costs are very high and so are transportation costs.
- Civil construction is costly due to erection of large towers.
- Wind turbine operation often leads to production of electronic noise which affects radar sites.
- In case of downwind HAWTs' the regular turbulence produced leads to structural failure.
- HAWTs require an additional yaw control mechanism to turn the blades toward the wind.

Types of HAWTs:**Mono-Blade Horizontal Axis Wind Turbine (HAWT)****Features:**

1. They have lighter rotor and are cheaper.
2. Blade are 15-25 m long and are made up of metal, glass reinforced plastics, laminated wood, composite carbon fiber/ fiberglass etc.
3. Power generation is within the range 15 kW to 50 kW and service life of plant is 30 years.

Advantages:

1. Simple and lighter construction.
2. Favorable price
3. Easy to install and maintain.

Disadvantages:

1. Tethering control necessary for higher loads.
2. Not suitable for higher power ratings.

Applications:

1. Field irrigation
2. Sea-Water desalination Plants
3. Electric power supply for farms and remote loads.

Twin-Blade HAWT

1. They have large sizes and power output in range of 1 MW, 2 MW and 3MW.
2. These high power units feed directly to the distribution network.

Blade HAWT

1. 3 blade propeller type wind turbines have been installed in India as well as abroad.
 2. The rotor has three blades assembled on a hub. The blade tips have a pitch control of 0 – 30° for controlling shaft speed.
 3. The shaft is mounted on bearings.
 4. The gear chain changes the speed from turbine shaft to generator shaft.
- Vertical axis Wind Turbines
- Vertical-axis wind turbines (or VAWTs) have the main rotor shaft arranged vertically as the plane of rotation is vertical. Blades are also vertical in this arrangement. The biggest advantage of VAWTs is they don't require a yaw control mechanism to be pointed into the wind. Thus these are useful in sites where wind direction is random or there is presence of large obstacles like trees, houses etc. Also VAWTs' don't require a tower structure and can be placed nearby a ground enabling access to electrical components. Some drawbacks are the low efficiency of wind production and the fact that large drag is created for rotating the blades in a vertical axis.

VAWT advantages

- A massive tower structure is not required, as VAWTs' are mounted closer to the ground
- They don't require yaw mechanisms.
- These are located closer to the ground and hence easier to maintain.
- These have lower startup speeds than their horizontal counterparts. These can start at speeds as low as 10Kmph.
- These have a lower noise signature.

VAWT disadvantages

- VAWTs' have lower efficiency as compared to HAWTs' because of the additional drag produced due to rotation of blades.
- Even though VAWTs' are located closer to the ground, the equipment now resides at the bottom of the turbines structure thus making it inaccessible.
- Because of their low height they cannot capture the wind energy stored in higher altitudes.

Types of VAWTs

Persian Windmill:

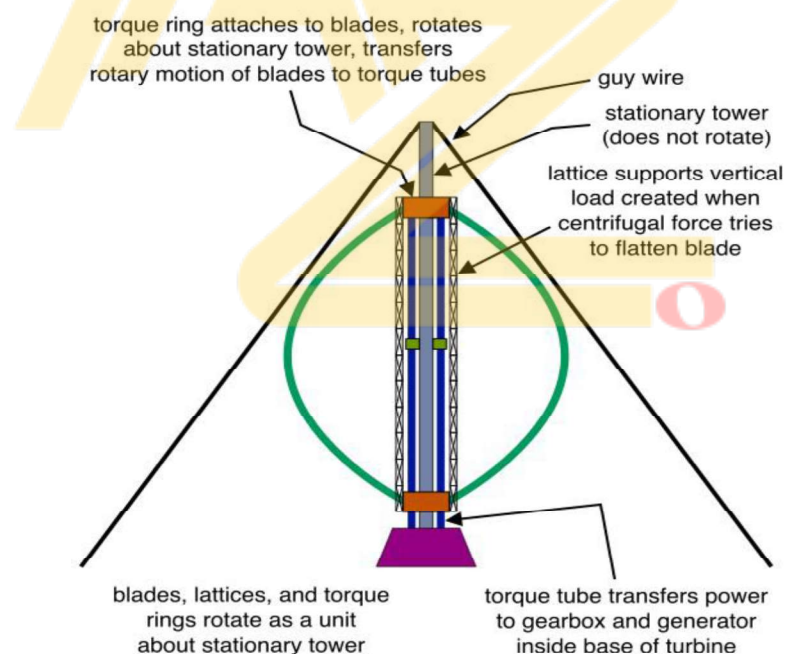
1. The Persian windmill was the earliest windmill installed. (7th Century A.D. – 13th Century A.D. in Persia, Afghanistan, and China)
2. It is a vertical axis windmill.. This windmill was used to grind grains and make flour.

Savonius Rotor VAWT:

1. Patented by S.J. Savonius in 1929.
2. It is used to measure wind current.
3. Efficiency is 31%.
4. It is Omni-directional and is therefore useful for places where wind changes direction frequently.

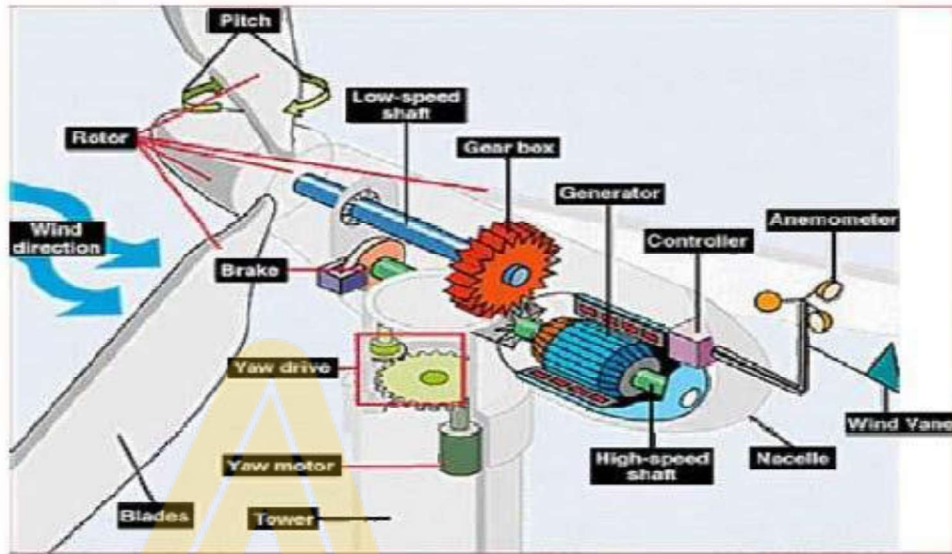
Darrieus Rotor VAWT:

1. It consists of 2 or 3 convex blades with airfoil cross-section.
2. The blades are mounted symmetrically on a vertical shaft.
3. To control speed of rotation mechanical brakes are incorporated. Those brakes consist of steel discs and spring applied air released calipers for each disc.



High Mechanical Efficiency Centrifugally Stable Darrieus Turbine

Internal Components of a Wind Turbine



Internal Components of a Wind Turbine

Anemometer: This device is used for measurement of speed. The wind speed is also fed to the controller as it is one of the variables for controlling pitch angle and yaw

Blades: These are aerodynamically designed structures such that when wind flows over them they are lifted as in airplane wings. The blades are also slightly turned for greater aerodynamic efficiency

Brake: This is either a mechanical, electrical or hydraulic brake used for stopping the turbine in high wind conditions

Controller: This is the most important part of the turbine as it controls everything from power output to pitch angle. The controller senses wind speed, wind direction, shaft speed and torque at one or more

Gear box: This step-up or steps down the speed of turbine and with suitable coupling transmits rotating mechanical energy at a suitable speed to the generator. Typically a gear box system steps up rotation speed from 50 to 60 rpm to 1200 to 1500 rpm

Generator: This can be a synchronous or asynchronous Ac machine producing power at 50Hz High-speed shaft: Its function is to drive the generator. Low-speed shaft: The rotor turns the low-speed shaft at about 30 to 60 rotations per minute.

Nacelle: The nacelle is the housing structure for high speed shaft, low speed shaft, gear box, generator, converter equipment etc. It is located atop the tower structure mostly in the shadow of the blades

Pitch: This is basically the angle the blades make with the wind. Changing the pitch angle changes whether the blades turn in or turn out of the wind stream.

Rotor: The hub and the blades together compose the rotor.

Tower: Towers are basically made up of tubular steel or steel lattice. Taller the towers greater is the amount of power generated as the wind speed generally goes on increasing with height. Also the temp of generator and power output produced is sensed

Wind direction: Generally erratic in nature, hence the rotor is made to face into the wind by means of control systems.

Wind vane: Basically the job of a wind sensor, measuring the wind speed and communicating the same to the yaw drive, so as to turn the turbine into the wind flow direction.

Yaw drive: This drive controls the orientation of the blades towards the wind. In case the turbine is out of the wind, then the yaw drive rotates the turbine in the wind direction

Yaw motor: Powers the yaw drive.

